1 Candidate code

FreeCAD #21257: 10 of the pieces the tracer finds

	piece of code	tokens
changed	Feature::getTopoShape	777
changed	Hole::execute	150
changed	ProfileBased::getProfileShape	232
callee	Hole::findHoles	668
callee	Hole::makeThread	1,561
callee	Measurement::getShape	99
caller	ProfileBased::getTopoShapeProfileWires	260
caller	CmdPartReverseShape::isActive	86
guide	CONTRIBUTING.md	2,221
test	TopoShapeMakeShape...Tests	285

2 Teacher LLM

holds the solution:
the PR's 6 human comments, R1–R6

supports
R1 R2
R3
–
R3
–
–
R4
–
–
–

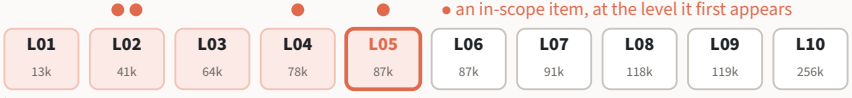
3 Exam scope

pieces some comment relies on, and the level where each piece first appears

in scope?	first level
in scope	L02
in scope	L02
not needed	L02
in scope	L04
not needed	L04
not needed	L04
in scope	L05
not needed	L05
not needed	L07
not needed	L09

4 From exam scope to a level

the label is the smallest nested level that contains every in-scope piece



label = L05 87k tokens, instead of 256k for L10

5 Training set

input: the PR, never the solution · target: its level

training PR (input x)	level y
FreeCAD #21257 Fix hole centered on point	L05
FreeCAD #19411 TechDraw: hard type enums	· · ·
. . . 196 PRs from 50 repositories	

Selector f_θ

learns PR $\mapsto$ level

pieces, tokens and levels are real; the teacher's links, and so the label, are illustrative